

The Complete Treatise on Econometrics in Human Resources Management

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Abstract

Human Resources Management (HRM) concerns a population of persons embedded in organizations through employment relationships. These relationships are simultaneously economic, legal, organizational, psychological, social, technological, and ethical. This paper develops a compact interdisciplinary framework for the econometrics of HRM, beginning with the measurement of group-specific employee flows and proceeding toward workforce composition, causal inference, survival analysis, machine learning, dynamic workforce systems, and responsible organizational decision-making.

The paper builds directly on three preceding contributions: a measurement framework for group-specific employee flows, a theoretical framework for their effects on workforce composition, and an exploratory data-science study of publicly observable employee-flow data. The first paper defines group-specific flow rates and distinguishes them from representation measures; the second establishes conditions under which absolute employee flows alter group representation; and the third demonstrates that real organizational data are heterogeneous, incomplete, and asymmetric in their reporting of hiring and separation information. These contributions motivate a broader econometric program.

The central proposition of this treatise is that increasing the granularity of HR measurement does not eliminate the employer–employee relationship. Rather, greater measurement resolution increases the capacity to study that relationship while preserving the ethical, legal, institutional, and human responsibilities attached to it. HR econometrics should therefore be understood as the study of employee and organizational states, flows, transitions, decisions, and outcomes under explicit assumptions about measurement, causality, uncertainty, and responsibility.

The treatise ends with “The End”

Contents

1	Introduction	3
2	The Three-Paper Foundation	3
2.1	Measurement	3
2.2	Theory	3
2.3	Exploratory Evidence	4
3	The Employer–Employee Relationship	4
4	The Workforce as a Dynamic System	5
5	Stocks, Flows, Rates, and Transitions	5
6	Descriptive Econometrics of HRM	6
7	Panel Data	6
8	Survival and Duration Analysis	6
9	Recruitment Econometrics	7
10	Compensation Econometrics	7

11 Promotion and Internal Labor Markets	7
12 Causal Econometrics	7
13 Measurement Error and Missingness	8
14 Organizational Heterogeneity	8
15 Psychology and Organizational Behaviour	8
16 Human Capital and Economics	9
17 Welfare Economics	9
18 Ethics and Philosophy	10
19 Religion and Moral Philosophy	10
20 Political Economy and International Relations	10
21 Warfare Economics and Organizational Resilience	11
22 Chemistry, Biology, and Systems Analogy	11
23 Statistics and Econometrics	11
24 Computer Science and Algorithms	11
25 Machine Learning in HRM	12
26 A General HR Decision Architecture	13
27 A Research Architecture for the Complete Treatise	13
28 Research Propositions	14
29 Empirical Research Agenda	14
30 A General Empirical Specification	15
31 The Central Logical Distinction	15
32 Conclusion	15

List of Figures

1 A stylized dynamic architecture of HR flows.	5
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List of Tables

1 Research architecture for HR econometrics.	14
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1 Introduction

Human Resources Management is frequently represented through administrative indicators: headcount, turnover, compensation, vacancies, absenteeism, performance, promotion, retention, and recruitment. Econometrics permits these observations to be connected through statistical models of relationships, transitions, counterfactuals, and dynamic systems.

The employment relationship, however, is not merely a statistical observation. The International Labour Organization describes it as a relationship between an employee and an employer under specified conditions and remuneration, giving rise to reciprocal rights and obligations [4, 5]. Consequently, an econometric description of HRM must not be confused with a complete account of the employment relationship.

The appropriate objective is therefore:

$$\begin{array}{c} HR \\ Econometrics \end{array} = \begin{array}{c} Measurement \\ + \\ Statistical \\ Inference \end{array} + \begin{array}{c} Causal \\ + \\ Identification \end{array} + \begin{array}{c} Organizational \\ + \\ Interpretation \end{array}. \quad (1)$$

A responsible framework additionally recognizes:

$$\text{Measurement precision} \neq \text{ethical permission}. \quad (2)$$

The employer remains responsible for decisions made within the employment relationship even when those decisions are supported by increasingly sophisticated statistical systems.

2 The Three-Paper Foundation

The present treatise begins from three preceding papers.

2.1 Measurement

Group-Specific Employee Flow Rates extends aggregate employee churn to identifiable employee groups. For group g over (t_1, t_2) , the proposed measure is

$$c_g(t_1, t_2) = \frac{h_g(t_1, t_2) - f_g(t_1, t_2)}{t_2 - t_1}, \quad (3)$$

where h_g and f_g denote group-specific hiring and firing counts [1].

The paper also distinguishes absolute group flow from representation,

$$p_g(t) = \frac{N_g(t)}{N(t)}, \quad (4)$$

and from relative flow,

$$q_g(t_1, t_2) = \frac{h_g(t_1, t_2) - f_g(t_1, t_2)}{N_g(t_1)(t_2 - t_1)}. \quad (5)$$

This distinction is fundamental because a flow, a stock, and a share of a stock are different statistical objects.

2.2 Theory

The second paper develops the theoretical implications of group-specific flow. Let

$$N(t) = \sum_{g=1}^G N_g(t). \quad (6)$$

For a fixed interval,

$$N_g(t_2) = N_g(t_1) + c_g \Delta t. \quad (7)$$

Aggregate employment satisfies

$$\frac{N(t_2) - N(t_1)}{\Delta t} = \sum_{g=1}^G c_g. \quad (8)$$

The representation of group g increases when

$$c_g > p_g(t_1) \sum_{j=1}^G c_j. \quad (9)$$

The right-hand side may be called the neutral-share flow,

$$c_g^* = p_g(t_1) \sum_{j=1}^G c_j, \quad (10)$$

and representation pressure may be defined as

$$\pi_g = c_g - c_g^*. \quad (11)$$

Thus,

$$\pi_g > 0 \Rightarrow p_g(t_2) > p_g(t_1), \quad (12)$$

$$\pi_g < 0 \Rightarrow p_g(t_2) < p_g(t_1), \quad (13)$$

$$\pi_g = 0 \Rightarrow p_g(t_2) = p_g(t_1). \quad (14)$$

The result establishes an important distinction between absolute growth and relative representation [2].

2.3 Exploratory Evidence

The third paper provides an empirical data-science layer. Its evidence layer contains 108 observations from seven organizations, with 55 hire observations, 48 turnover observations, four employment-stock observations, and one other-separation observation. Directly observed group-specific firing observations are absent [3].

This absence must not be interpreted as zero firing. Formally,

$$\text{not reported} \neq 0. \quad (15)$$

This is a general principle of HR data science: missingness is itself part of the measurement problem.

3 The Employer–Employee Relationship

The employment relationship has economic and legal structure. The employee supplies labor, skills, knowledge, time, and organizational participation; the employer supplies remuneration and organizational coordination and assumes corresponding responsibilities.

A simplified relationship can be represented as

$$\mathcal{R}_{it} = (L_{it}, W_{it}, C_{it}, P_{it}, T_{it}, D_{it}), \quad (16)$$

where L denotes labor supplied, W remuneration, C contractual conditions, P organizational participation, T rights and obligations, and D decision structures.

Econometric analysis observes only projections of this multidimensional relationship.

$$Y_{it} = \phi(\mathcal{R}_{it}, X_{it}, \varepsilon_{it}). \quad (17)$$

Consequently, statistical observability must not be confused with ontological completeness.

4 The Workforce as a Dynamic System

Let the workforce contain G groups. Define

$$\mathbf{N}_t = \begin{pmatrix} N_{1t} \\ N_{2t} \\ \vdots \\ N_{Gt} \end{pmatrix}. \quad (18)$$

A more complete employee-flow identity distinguishes hiring, voluntary separation, and employer-initiated separation:

$$\mathbf{N}_{t+1} = \mathbf{N}_t + \mathbf{H}_t - \mathbf{S}_t - \mathbf{F}_t. \quad (19)$$

Thus,

$$\mathbf{C}_t = \mathbf{H}_t - \mathbf{S}_t - \mathbf{F}_t. \quad (20)$$

The representation vector is

$$\mathbf{p}_t = \frac{\mathbf{N}_t}{\mathbf{1}'\mathbf{N}_t}. \quad (21)$$

This formulation converts workforce composition into a dynamic population problem. The relevant mathematical ideas have analogues in population dynamics, physics, systems theory, biology, and Markov processes.

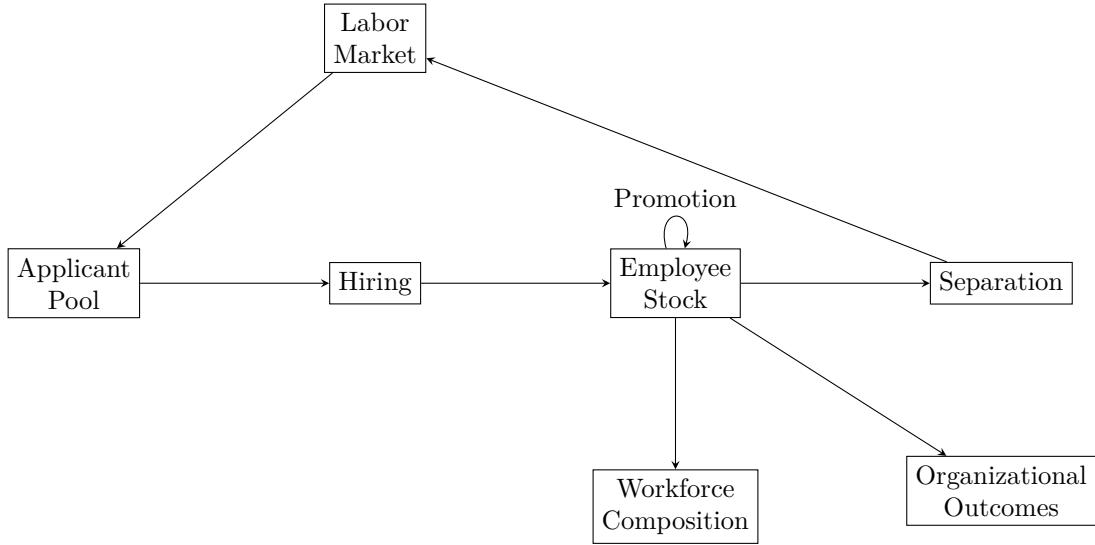


Figure 1: A stylized dynamic architecture of HR flows.

5 Stocks, Flows, Rates, and Transitions

HR econometrics should distinguish at least four classes of quantities:

1. stocks;
2. flows;
3. rates;
4. transition probabilities.

A stock is measured at a point in time. A flow is measured over an interval. A rate normalizes a flow by time or population. A transition probability describes movement between states.

For example, let an employee occupy state $s \in \mathcal{S}$. Then

$$P_{ab,t} = \Pr(S_{i,t+1} = b \mid S_{it} = a, X_{it}). \quad (22)$$

A workforce may therefore be represented by a transition matrix

$$\mathbf{P}_t = \begin{pmatrix} p_{11,t} & \cdots & p_{1G,t} \\ \vdots & \ddots & \vdots \\ p_{G1,t} & \cdots & p_{GG,t} \end{pmatrix}. \quad (23)$$

This connects HRM to Markov processes, survival analysis, operations research, population biology, and dynamical systems.

6 Descriptive Econometrics of HRM

The first econometric stage is descriptive.

For an outcome Y_{it} ,

$$Y_{it} = \alpha + \beta' X_{it} + \varepsilon_{it}. \quad (24)$$

The coefficient β describes conditional association under the specified model. It does not automatically represent a causal effect.

For group-specific employee flow,

$$c_{git} = \alpha_g + \lambda_t + \beta' X_{git} + u_{git}. \quad (25)$$

Organization effects may be introduced through

$$c_{git} = \alpha_i + \gamma_g + \lambda_t + \beta' X_{git} + u_{git}. \quad (26)$$

This immediately raises the classical econometric problems of omitted variables, heterogeneity, selection, measurement error, simultaneity, and dependence.

7 Panel Data

Organizations and employees naturally generate panel data.

A two-way fixed-effects model can be written as

$$Y_{igt} = \alpha_i + \gamma_g + \lambda_t + \beta' X_{igt} + \varepsilon_{igt}. \quad (27)$$

Here α_i absorbs time-invariant organizational heterogeneity, γ_g absorbs persistent group characteristics, and λ_t captures common time shocks.

Random-effects models may instead be appropriate when the assumptions required for uncorrelated latent heterogeneity are defensible. The distinction is empirical rather than ideological [13].

8 Survival and Duration Analysis

Employee retention is inherently a duration problem.

Let T_i denote the duration of employment. The survival function is

$$S(t) = \Pr(T_i > t), \quad (28)$$

and the hazard function is

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}. \quad (29)$$

A Cox proportional-hazards specification is

$$h_i(t) = h_0(t) \exp(\beta' X_i(t)). \quad (30)$$

Competing-risks models become appropriate when employees may leave through distinct routes such as resignation, retirement, dismissal, or redundancy.

9 Recruitment Econometrics

Recruitment can be treated as a sequence of conditional choices.

For candidate i and job j ,

$$\Pr(Y_{ij} = 1 \mid X_{ij}) = \Lambda(\beta' X_{ij}), \quad (31)$$

where Λ is an appropriate link function.

A multinomial formulation may represent alternatives such as

$$Y_i \in \{\text{reject, offer, accept, alternative employment}\}. \quad (32)$$

Selection occurs at multiple stages. Consequently, an observed hiring sample is not necessarily representative of the original applicant population.

10 Compensation Econometrics

A basic earnings equation is

$$\ln W_{it} = \alpha + \beta' X_{it} + \gamma_i + \lambda_t + \varepsilon_{it}. \quad (33)$$

Potential explanatory variables include education, experience, occupation, tenure, location, skills, organizational characteristics, bargaining institutions, and job responsibilities.

The econometric interpretation must distinguish descriptive wage differences from causal effects and from normative judgments concerning fairness.

11 Promotion and Internal Labor Markets

Promotion can be modelled as a transition:

$$P_{it}^{\text{prom}} = \Pr(Y_{it} = 1 \mid X_{it}, Z_{it}). \quad (34)$$

A dynamic internal labor market can then be represented as

$$\text{Entry} \rightarrow \text{Development} \rightarrow \text{Promotion} \rightarrow \text{Retention} \rightarrow \text{Exit}. \quad (35)$$

This connects HR econometrics with personnel economics, human-capital theory, labor economics, organizational behaviour, and career theory [9, 7].

12 Causal Econometrics

Causal inference requires a counterfactual.

Let $Y_i(1)$ denote the outcome under treatment and $Y_i(0)$ the outcome without treatment. The individual treatment effect is

$$\tau_i = Y_i(1) - Y_i(0). \quad (36)$$

The average treatment effect is

$$ATE = E[Y_i(1) - Y_i(0)]. \quad (37)$$

The fundamental problem is that both potential outcomes are not simultaneously observable for the same individual [10, 11, 12].

HR examples include:

- introducing a recruitment technology;
- changing compensation;
- introducing remote work;
- restructuring a department;
- implementing training;
- adopting AI-assisted management.

Difference-in-differences may be written as

$$Y_{it} = \alpha_i + \lambda_t + \beta(D_i \times Post_t) + \varepsilon_{it}. \quad (38)$$

The interpretation of β depends on the identifying assumptions, especially the counterfactual trend structure.

13 Measurement Error and Missingness

The empirical HR researcher must distinguish the latent quantity from its measurement.

Let

$$Y_{it}^* = Y_{it} + u_{it}, \quad (39)$$

where Y^* is the observed measure and u is measurement error. Missingness may be represented through an indicator

$$R_{it} = \begin{cases} 1, & \text{if } Y_{it} \text{ is observed,} \\ 0, & \text{otherwise.} \end{cases} \quad (40)$$

The missingness mechanism may be:

$$R \perp Y \quad (\text{MCAR}), \quad (41)$$

$$R \perp Y \mid X \quad (\text{MAR}), \quad (42)$$

$$R \not\perp Y \mid X \quad (\text{MNAR}). \quad (43)$$

The third foundational paper demonstrates why this issue is particularly important in HR: publicly reported organizational data can differ substantially in concept, unit, group dimension, and period [3].

14 Organizational Heterogeneity

An organization is not a homogeneous statistical unit.

Let organizational outcomes depend on:

$$Y_{it} = f(\text{industry, size, technology, geography, management, labor market, institutions}). \quad (44)$$

Cross-organizational comparisons therefore require careful harmonization.

The empirical principle should be:

$$\text{source quantity} \rightarrow \text{validated observation} \rightarrow \text{comparable subset}, \quad (45)$$

rather than silently transforming heterogeneous observations into a supposedly common construct [3].

15 Psychology and Organizational Behaviour

Econometric HRM cannot be reduced to observable economic incentives.

Employees possess preferences, beliefs, identities, expectations, perceptions, emotions, social relationships, and cognitive limitations. Relevant constructs include:

- motivation;
- organizational commitment;
- job satisfaction;
- perceived organizational support;
- psychological safety;
- trust;
- identity;
- social comparison;
- stress;
- autonomy.

A generalized behavioural model may be represented as

$$Y_{it} = f(I_{it}, M_{it}, S_{it}, R_{it}, C_{it}, X_{it}, \varepsilon_{it}), \quad (46)$$

where I denotes incentives, M motivation, S social context, R relationships, and C organizational conditions.

Psychological variables should be measured rather than inferred casually from administrative outcomes.

Clinical psychiatric diagnosis is outside the scope of ordinary HR econometrics. Workplace distress and mental-health-related outcomes should be studied through appropriate validated instruments, privacy safeguards, professional standards, and applicable law rather than through speculative inference.

16 Human Capital and Economics

Human-capital theory treats education, skills, experience, and knowledge as productive resources [7, 8].

For a worker i , a stylized human-capital stock may be written

$$K_{it} = K_{i,t-1} + I_{it} - D_{it}, \quad (47)$$

where I represents investment and D depreciation or obsolescence.

At the organizational level,

$$K_t = \sum_{i=1}^{N_t} K_{it}. \quad (48)$$

Employee flows therefore alter not only headcount but potentially the organization's productive knowledge stock.

17 Welfare Economics

HR decisions may create benefits and costs for multiple parties.

Let social welfare be

$$W = \sum_i u_i + \theta \Pi - C, \quad (49)$$

where u_i represents employee utility, Π organizational surplus, and C external or social costs.

The parameter θ represents the institutional weight assigned to organizational surplus relative to employee welfare.

This formulation makes explicit that organizational optimization and social welfare optimization need not be identical.

18 Ethics and Philosophy

The econometric treatment of HRM involves at least three distinct questions:

1. What happened?
2. What caused it?
3. What ought to be done?

These are not interchangeable.

Descriptive statistics address the first question. Causal econometrics addresses the second under explicit assumptions. Ethics and political philosophy address the third.

This distinction prevents the common logical error

$$\text{observed difference} \not\Rightarrow \text{causal discrimination} \not\Rightarrow \text{normative conclusion}. \quad (50)$$

The three foundational papers explicitly preserve this distinction [1, 2].

19 Religion and Moral Philosophy

Religious and philosophical traditions can supply normative vocabularies concerning duty, dignity, reciprocity, justice, responsibility, stewardship, and the treatment of persons. Such traditions should not be substituted for empirical identification.

A general principle can nevertheless be stated:

$$\text{Person} \neq \text{mere statistical observation}. \quad (51)$$

The statistical abstraction of an employee as i is analytically useful but does not exhaust the employee's moral or social status.

This distinction is compatible with diverse traditions, including Aristotelian ethics, Kantian ethics, utilitarianism, virtue ethics, and religious traditions emphasizing duty, stewardship, compassion, justice, and human dignity.

20 Political Economy and International Relations

HRM occurs within political and institutional environments.

A multinational organization may operate under different:

- labor laws;
- tax regimes;
- social-insurance systems;
- collective-bargaining institutions;

- migration regimes;
- data-protection regimes;
- industrial policies;
- geopolitical constraints.

Thus,

$$Y_{ict} = f(X_{ict}, L_{ct}, I_{ct}, G_{ct}, \varepsilon_{ict}), \quad (52)$$

where c denotes country, L legal institutions, I labor-market institutions, and G geopolitical conditions.

International comparison therefore requires institutional rather than merely numerical harmonization.

21 Warfare Economics and Organizational Resilience

Periods of war, sanctions, pandemics, natural disasters, or geopolitical disruption can alter labor supply and organizational demand simultaneously.

A stylized shock model is

$$\mathbf{N}_{t+1} = \mathbf{N}_t + \mathbf{C}_t + \mathbf{S}_t^{ext}, \quad (53)$$

where $\mathbf{S}^{ext}$ represents an exogenous labor-market shock.

Resilience can be represented by

$$R_t = \frac{\text{critical workforce capacity retained}}{\text{critical workforce capacity required}}. \quad (54)$$

The purpose of such a formulation is not to militarize HRM, but to recognize that organizations operate within economic systems that can experience severe external shocks.

22 Chemistry, Biology, and Systems Analogy

The use of interdisciplinary analogies must remain disciplined.

Biology provides population dynamics, birth-death processes, selection, adaptation, and competition. Chemistry provides reaction systems and rate concepts. Physics provides state-space descriptions, conservation identities, equilibrium, flows, and dynamical systems.

An HR workforce may be represented abstractly as a population:

$$\frac{dN_g}{dt} = H_g(t) - S_g(t) - F_g(t). \quad (55)$$

This resembles a population balance equation, but the analogy does not imply that employees are biological particles or chemical reactants. Human agency remains a fundamental distinction.

23 Statistics and Econometrics

The econometric hierarchy can be summarized as

$$\begin{aligned} & \text{Observation} \rightarrow \text{Measurement} \rightarrow \text{Description} \rightarrow \text{Association} \rightarrow \text{Identification} \rightarrow \begin{matrix} \text{Causal} \\ \text{Inference} \end{matrix} \rightarrow \\ & \text{Decision.} \end{aligned} \quad (56)$$

Each arrow requires additional assumptions.

The statistical model

$$Y = X\beta + \varepsilon \quad (57)$$

does not become causal merely because β is statistically significant.

Inference requires consideration of sampling, specification, dependence, heteroskedasticity, clustering, multiple testing, model uncertainty, and external validity [13, 12].

24 Computer Science and Algorithms

HR analytics increasingly operates through computational pipelines.

A reproducible pipeline can be represented as follows.

INPUT:

- organizational records
- employee records
- recruitment records
- compensation records
- external labor-market data

VALIDATE:

- schema
- provenance
- units
- dates
- group definitions
- missingness

TRANSFORM:

- construct stocks
- construct flows
- construct rates
- construct representations

ANALYZE:

- descriptive statistics
- panel models
- survival models
- causal models
- machine-learning models

ROBUSTNESS:

- alternative specifications
- sensitivity analysis
- subgroup analysis
- missingness analysis

OUTPUT:

- estimates
- uncertainty
- diagnostics
- reproducible records
- decision-relevant evidence

The computational principle is:

$$\text{Reproducibility} = \text{Data Provenance} + \text{Code} + \text{Specification} + \text{Environment}. \quad (58)$$

25 Machine Learning in HRM

Machine learning may be used for prediction without being confused with causal inference.

For a prediction target Y ,

$$\hat{Y}_{t+1} = f_{\theta}(X_t). \quad (59)$$

Potential applications include:

- turnover prediction;
- workforce-demand forecasting;
- recruitment forecasting;
- vacancy duration prediction;
- skill-demand prediction;
- workforce composition forecasting.

The objective function might be

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n L(y_i, f_{\theta}(x_i)). \quad (60)$$

However,

$$\text{predictive accuracy} \neq \text{causal validity}. \quad (61)$$

AI-enabled HR systems also create governance questions concerning validity, transparency, privacy, accountability, explainability, and harmful bias. The NIST AI Risk Management Framework explicitly organizes trustworthy AI around such concerns [16]. OECD research similarly identifies worker trust, training, consultation, privacy, autonomy, and job quality as important dimensions of workplace AI [17, 18].

26 A General HR Decision Architecture

The complete analytical system can be represented as

$$\mathcal{H} = (\mathcal{D}, \mathcal{M}, \mathcal{E}, \mathcal{C}, \mathcal{A}, \mathcal{R}), \quad (62)$$

where:

- $\mathcal{D}$ = data;
- $\mathcal{M}$ = measurement;
- $\mathcal{E}$ = econometric estimation;
- $\mathcal{C}$ = causal analysis;
- $\mathcal{A}$ = algorithmic prediction;
- $\mathcal{R}$ = responsibility and governance.

A decision rule can then be represented as

$$a^* = \arg \max_{a \in \mathcal{A}} E[U(a, Y) \mid D], \quad (63)$$

subject to legal, ethical, organizational, and operational constraints.

The constraints matter:

$$a \in \mathcal{A}_{\text{legal}} \cap \mathcal{A}_{\text{ethical}} \cap \mathcal{A}_{\text{organizational}}. \quad (64)$$

Thus the econometric optimum is not necessarily the permissible organizational decision.

27 A Research Architecture for the Complete Treatise

The research program can be organized into the following sequence.

Stage	Object	Principal Methods
I	Workforce measurement	Stocks, flows, rates, representations
II	Employee-flow theory	Flow decomposition, dynamic identities
III	Exploratory HR data science	Provenance, missingness, comparability
IV	Descriptive econometrics	Regression, GLM, panel data
V	Workforce dynamics	Markov, state-space, transition models
VI	Employee duration	Survival and competing risks
VII	Causal HR econometrics	DiD, IV, RDD, matching, experiments
VIII	Human capital	Wages, skills, productivity, promotion
IX	Organizational behaviour	Motivation, trust, commitment, culture
X	AI/ML	Prediction, causal ML, algorithmic management
XI	Governance	Privacy, fairness, accountability, explainability
XII	Strategic HRM	Workforce planning, resilience, organizational design

Table 1: Research architecture for HR econometrics.

28 Research Propositions

The three foundational papers imply several propositions.

Proposition 1. For a fixed positive interval, positive group-specific net flow implies an increase in absolute group employment.

Proposition 2. Positive absolute group flow does not imply positive representation change.

Proposition 3. During aggregate workforce contraction, negative group flow does not necessarily imply declining representation.

Proposition 4. Equal net employee flows do not imply equal gross hiring and separation flows.

Proposition 5. Differences in group-specific employee flows are descriptive evidence of different flows, not by themselves evidence of discriminatory intent.

The first four propositions follow from the theoretical structure developed in the second paper; the fifth follows from the methodological boundary established in the first two papers [1, 2].

29 Empirical Research Agenda

The current evidence suggests several immediate research priorities:

1. expand the number of organizations;
2. obtain multi-year observations;
3. increase sectoral coverage;
4. obtain group-specific employer-initiated separation measures;
5. distinguish voluntary from involuntary turnover;
6. harmonize definitions without erasing source differences;
7. construct longitudinal organizational panels;
8. estimate dynamic flow models;
9. test causal mechanisms;
10. develop reproducible computational pipelines.

The exploratory study specifically identifies Canada, Australia, manufacturing, healthcare, retail, telecommunications, energy, transportation, consumer goods, multi-year reporting, and employer-initiated separation measures as important areas for future acquisition [3].

30 A General Empirical Specification

A general group-level panel model may be written as

$$c_{git} = \alpha_i + \gamma_g + \lambda_t + \beta' X_{git} + \delta' Z_{it} + u_{git}. \quad (65)$$

A representation equation may be written as

$$p_{git} = \frac{N_{git}}{\sum_j N_{jit}}. \quad (66)$$

A dynamic model can then take the form

$$p_{g,i,t+1} = f(p_{g,it}, c_{git}, \mathbf{c}_{it}, X_{git}, Z_{it}, \varepsilon_{git}). \quad (67)$$

The econometric research question becomes not merely whether a group changes, but which components of the organizational system explain the change.

31 The Central Logical Distinction

The entire treatise can be summarized by three logical propositions.

First,

$$\text{measurement} \Rightarrow \text{description}. \quad (68)$$

Second,

$$\text{description} \not\Rightarrow \text{causation}. \quad (69)$$

Third,

$$\text{causal evidence} \not\Rightarrow \text{normative conclusion}. \quad (70)$$

The final step belongs to ethics, law, organizational governance, and human judgment.

Conversely, ethical commitments cannot substitute for empirical analysis when the question concerns what actually happened.

Thus,

$$\boxed{\text{Good HRM} = \text{Evidence} + \text{Causality} + \text{Human Responsibility}.} \quad (71)$$

32 Conclusion

The econometrics of Human Resources Management can be developed as a unified field whose object is the dynamic employer–employee system.

The three foundational papers establish the initial sequence:

$$\text{Measurement} \rightarrow \text{Theory} \rightarrow \text{Exploratory Evidence}. \quad (72)$$

The present treatise extends that sequence:

$$\begin{aligned} \text{Measurement} \rightarrow \text{Theory} \rightarrow \text{Exploration} \rightarrow \text{Econometrics} \rightarrow \begin{matrix} \text{Causal} \\ \text{Inference} \end{matrix} \rightarrow \text{Prediction} \rightarrow \\ \begin{matrix} \text{Responsible} \\ \text{Decision} \end{matrix} \end{aligned} \quad (73)$$

The fundamental contribution is therefore not a single estimator. It is an analytical architecture.

Employees can be represented as observations without being reduced to observations. Organizations can be represented as systems without eliminating individual agency. Workforce flows can be measured without presuming their causes. Statistical differences can be investigated without automatically converting them into moral conclusions. Machine learning can improve prediction without becoming causal

inference. And econometric sophistication can increase without diminishing the responsibilities created by the employment relationship.

The ultimate object of HR econometrics is therefore neither the employee nor the organization considered in isolation, but the measurable dynamics of the relationship between them.

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Glossary

Attrition

Loss of employees from an organization, with the exact meaning determined by the source or organizational definition.

Causal inference

Statistical reasoning concerning counterfactual changes attributable to an intervention or treatment under explicit identifying assumptions.

Employee flow

Movement of employees into, within, or out of an organization over an interval.

Employee stock

Number of employees belonging to a specified population at a particular point in time.

Employer-initiated separation

An employee departure initiated by the employer, subject to the definition used in the underlying data.

Group representation

The proportion of total organizational employment belonging to a specified group.

Group-specific employee flow rate

Net employee flow for a specified group divided by the length of the observation interval.

Hazard

The instantaneous conditional rate of occurrence of an event, such as employee exit.

Human capital

Skills, knowledge, experience, and other productive capabilities embodied in workers.

Machine learning

Computational methods that estimate predictive or decision functions from data.

Measurement error

Difference between an observed quantity and the latent quantity it is intended to measure.

Missingness

Non-observation of a variable or record. Missingness should not automatically be interpreted as a zero value.

Net employee flow

The difference between relevant inflows and outflows over a specified interval.

Neutral-share flow

The group-specific flow that would preserve the group’s existing representation in the workforce.

Panel data

Observations on multiple entities across multiple time periods.

Representation pressure

The difference between actual group flow and the neutral-share flow.

Survival analysis

Statistical analysis of time until an event occurs.

Turnover

Employee movement into and/or out of an organization according to the definition adopted by the data source.

Workforce composition

The distribution of employees across specified groups or categories.

Workforce dynamics

The evolution of workforce stocks, flows, transitions, and composition through time.

The End